Unit 2 Poster — Claim, Proof, Repair

Period B · Tue 11/10 + Thu 11/12 build, Fri 11/13 gallery walk · CED topics 2.1 to 2.9 (nothing binomial) · one DOK-3 claim per group

Printables (PDF): group sheets (six cards, two pages each; print one per group) · example board (project this first) · gallery-walk sheet (one per student, Friday) · teacher key (this guide, the board, the cards with answers, exit tickets, run of show, observer card, question and hint cards, rubric and scoring grid, answer key).

The hook for students: "Every claim on these cards is one somebody could really make about data we have already used. Three of them come straight from questions this class asked in the last two weeks. Your group's job is a verdict, the proof, and a repair. Then your poster teaches the room."

The six cards

#	Topics	The claim being judged	The mistake it busts (Period B evidence)	Data
1	2.1, 2.2	"Freeze Time people want to be rich: 26 chose Rich, more than any other power."	"The bigger count wins" **PERSISTENT B**: Period B's Unit 1 error, and 3 students read a relative-frequency bar chart as counts on quiz 2.1 (10/29).	Super Status table, 433 students
2	2.6	"Most students who want to be Famous chose Fly, so most students who chose Fly want to be Famous."	P(A | B) read as P(B | A). Asked in class 10/30: "why do the two conditional probabilities differ?"	Super Status table
3	2.5, 2.7	"Nobody who chose Telepathy wants to be Famous (0 of 68), so Telepathy and Famous are independent."	Mutually exclusive confused with independent. Asked 10/29; you promised the class an example of mutually exclusive events that are not independent. **PERSISTENT B**	Super Status table + a 2x2 table the group designs
4	2.4, 2.7	"P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%, and grads are no more likely than anyone else to get news online: only 9.8% are grads who read news online."	Addition rule without the subtraction; a joint probability used where a conditional belongs. 2.7 is B's least-finished lesson (worksheet mean 46; 8 of 14 not done).	2010 AP exam table, 2,500 adults (2.7 follow-along)
5	2.3, 2.8, 2.9	"A fair game! You win on 2 of every 6 rolls."	Expected value read as "every game"; winnings used instead of net; "fair" without asking for whom (2.8 exit-ticket traps). Also the 10/30 TI-84 promise.	Carnival die game (2.8 follow-along exit ticket)
6 spare	2.3	"6 heads in a row in 20 flips is impossible for a fair coin. It's rigged."	A simulation estimates; it never proves one outcome impossible (10/26: "was the simulated coin fair?").	randInt(0,1,20) on the TI-84

Coverage: 2.1 and 2.2 on card 1; 2.3 on cards 5 and 6; 2.4 everywhere (probabilities as relative frequencies, the complement region on card 4); 2.5 on card 3; 2.6 on cards 1, 2 and 4; 2.7 on cards 3 and 4; 2.8 and 2.9 on card 5. The walk sheet makes every student compute one number at each poster except their own, so everyone meets every topic.

Unit 2 Poster — groups, roles, days, materials

Groups

Class size	Groups	Who gets what
14 (Period B)	4 trios + 1 pair	The pair takes card 2 (two probabilities, the lightest build). A strong trio takes card 4 (the lesson most of the class has not finished) and another takes card 3 (the proof). Card 5 wants one calculator-confident student. Card 6 is the spare for a sixth group or early finishers.
12-13	4 groups	Drop card 1; its segmented bar chart becomes the Friday debrief demo.
15-18	6 groups	Card 6 goes to the sixth group.

Roles inside a trio

The whole group judges the claim together **first**, on scratch paper and whiteboards, before anyone touches the poster. Then:

- **Table Keeper** — every fraction and percent on the poster, with the arithmetic shown and the denominator named.
- **Visual** — the diagram the card asks for (segmented bar chart, shaded tables, Venn, 2x2 table, dotplot): title, labels, a key or scale.
- **Writer** — the verdict sentence, the repaired claim and the explanation sentences, in context, using the words from the language objective; checks the must-have list before the walk.

In a pair, the Table Keeper also draws the visual.

The three days (details on the run-of-show pages)

Tue 11/10 · 65 min · build	Thu 11/12 · 65 min · observed: defend, repair, finish	Fri 11/13 · 55 min · gallery walk
Do Now on boards. Objectives + EQ posted. Three Reads on the shared table. Example board. Cards out. Solve verdict + proof on scratch paper first, then start the poster. Exit ticket 1.	Do Now on boards. Defend round with question cards (poster 1 with 2, 3 with 4; poster 5 presents to you). Repair part of each card. Finish against the checklist. Five minutes on the 10/29 promise. Exit ticket 2, alone, on paper.	Final touches, posters up. Walk sheet at every other poster. Debrief on the two persistent ones. Score with the grid.

Materials and grading

Six poster boards (one spare), markers (four colors per group), rulers, tape, scratch paper, a whiteboard + marker + eraser for every student, TI-84s (at least two for card 5, ideally one per group), the printed cards (two pages, one per group), 14 walk sheets, 14 of each exit ticket, one question card per group, the hint cards, the board page (projected or printed large), the example board (projected). Each card's five must-haves are the rubric: all five = E, three or four = P, fewer = I. Scores E 115 / P 100 / I 90, any integer 90 to 115 per sheet; enter them with `node scripts/enter-poster.mjs` (format on the rubric page). Posters are 30% of Q2 Work for Period B.

AP Statistics · Period B · Unit 2 Poster

Tue 11/10 · Thu 11/12 · Fri 11/13 **DOK 3** judge it · prove it · repair it

Content objective

I can judge a claim about chance: find the number it really depends on, prove my verdict with it, and repair the claim so it is true.

Language objective

I can defend my verdict out loud and in writing, in complete sentences, using the word my claim needs: **given, union, independent, mutually exclusive** or **expected value**.

Essential question

What makes a claim about chance worth believing?

Frames

Verdict: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."
Partner talk: "I agree / disagree with _____ because _____." "Say it again with your key word."
Poster 5: "In the long run, _____ per _____."

Key words: given | union | independent | mutually exclusive | expected value **Helpers:** denominator | long run

This week

- **Tuesday:** verdict and proof on scratch paper first. Then the poster.
- **Thursday:** defend your verdict to another group. Repair the claim. Finish against the checklist.
- **Friday:** gallery walk. One number at each poster.

Poster 1 — Who wants to be rich?

DOK 3

Data

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. For each superpower, what percent of its fans chose Rich? Write a fraction and a percent. Which total is the denominator, and why?

> **Answer.** Fly 22/89 = 24.7% · Freeze Time 26/101 = 25.7% · Invisibility 22/98 = 22.4% · Super Strength 13/77 = 16.9% · Telepathy 19/68 = 27.9%. The denominator is the row total: "given Fly" means only the 89 Fly students count. Overall, P(Rich) = 102/433 = 23.6%.

b. Draw a segmented bar chart: one 100% bar per superpower, split into Famous / Happy / Healthy / Rich. Whole percents are fine. Add a key and a percent axis.

> **Answer** (percents within each superpower: Famous / Happy / Healthy / Rich). Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9. Overall 3.5 / 61.2 / 11.8 / 23.6.

c. Verdict. Do Freeze Time fans stand out for wanting to be rich? Compare their percent with the other four groups and with all 433 students.

> **Answer.** No. The count is right (26 is the largest Rich count), but Freeze Time is also the largest group (101 fans), so a large count is expected. By percent Freeze Time is 25.7%, close to the 23.6% of all students. Telepathy is highest at 27.9% (Freeze Time is second, 2.2 points behind), and every group sits between 17% and 28%. No group stands out: not supported. Also accept the true reading of the count: of the 102 students who want Rich, 26 (25.5%) chose Freeze Time, more than any other power. That is P(Freeze Time | Rich), not the claim's P(Rich | Freeze Time).

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

> **Answer** (any true version). "About 1 in 4 students in every superpower group wants to be rich (17% to 28%); Freeze Time fans, at 25.7%, are close to the 23.6% of all students." Also true: "Among students who chose Telepathy, 27.9% want to be rich, the highest share; Freeze Time is second at 25.7%."

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

> **Answer.** Yes: the conditional distributions differ. Telepathy fans: 0% Famous, 0% Healthy, 72% Happy. Fly fans: 5.6% Famous, 15.7% Healthy, 54% Happy. With no association every bar would be split the same way.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 1 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Five "Rich" percents, each with its fraction
- ☐ Segmented bar chart: five bars, each 100%, four segments, a key, a percent axis
- ☐ A verdict that compares Freeze Time's percent with the other groups and with all 433 students
- ☐ The repaired claim, true, with a percent
- ☐ One association sentence comparing two superpowers' bars

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The five Rich fractions and percents, then the percents for the bars.	The segmented bar chart: five 100% bars, four segments, key, percent axis, title.	The verdict, the repaired claim, the association sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given · denominator · association

Draft the poster sentences here. Every sentence names the context and uses a number from the table.

Poster 2 — The flipped headline

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Find P(Fly | Famous) and P(Famous | Fly) as fractions and percents. Which total is the denominator each time, and why?

> **Answer.** P(Fly | Famous) = 5/15 = 33.3%: the 15 students who want to be Famous (a column total). P(Famous | Fly) = 5/89 = 5.6%: the 89 students who chose Fly (a row total). Same numerator, different "given" group.

b. Shade the "given" group on two copies of the table, one for each probability. Circle the cell the two share.

> **Answer.** Copy 1: the Famous column shaded (total 15) for P(Fly | Famous). Copy 2: the Fly row shaded (total 89) for P(Famous | Fly). The shared cell is 5: same numerator, different given group.

c. Verdict on each half. Is "most" right in the first sentence? Does the second sentence follow from the first?

> **Answer.** First half: Fly is the top pick among the 15 Famous students (5; next is Super Strength with 4), but 33% is not "most": a plurality, not a majority. Second half: it does not follow; the condition is flipped. Only 5.6% of Fly-choosers want to be Famous.

d. Repair. Rewrite the headline as two true sentences, each with its probability.

> **Answer.** "Among the 15 students who want to be Famous, Fly was the most popular power (5, or 33%). Among the 89 who chose Fly, only 5.6% want to be Famous."

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

> **Answer** (any correct pair). P(Happy | Super Strength) = 43/77 = 55.8% vs P(Super Strength | Happy) = 43/265 = 16.2%. P(Happy | Freeze Time) = 63/101 = 62.4% vs P(Freeze Time | Happy) = 63/265 = 23.8%. P(Rich | Telepathy) = 19/68 = 27.9% vs P(Telepathy | Rich) = 19/102 = 18.6%. Why: the numerator is the same cell; the denominator is the size of the given group.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 2 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Both conditional probabilities as fractions and percents
- ☐ Two copies of the table, each with its "given" group shaded
- ☐ A verdict on each half of the headline, each with its number
- ☐ The repaired headline: two true sentences with their numbers
- ☐ A second flipped pair from the table with both probabilities and a one-sentence why

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The two fractions and percents, and the second flipped pair. Name each denominator.	The two shaded tables, with the shared cell circled.	The verdict on each half, the repaired headline, the why sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given denominator

Draft the poster sentences here.

Poster 3 — Can't happen together, so unrelated?

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"*Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other.*"

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

> **Answer.** P(Telepathy and Famous) = 0/433 = 0. No student in this table did both, so for a student picked from this table the two events are mutually exclusive. (The 0 shows it did not happen among these 433, not that it never could.)

b. Are they independent? Compare P(Famous) with P(Famous | Telepathy).

> **Answer.** P(Famous) = 15/433 = 3.5% but P(Famous | Telepathy) = 0/68 = 0%: not equal, so not independent. (The product check agrees: P(Famous) × P(Telepathy) = (15/433)(68/433) = 0.0054, but P(Famous and Telepathy) = 0.)

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

> **Answer.** "Telepathy and Famous are mutually exclusive but not independent: knowing a student chose Telepathy drops the chance they want to be Famous from 3.5% to 0." Seeing one rules the other out, which changes its probability: that is dependence, not independence.

d. Prove it in general. Two events can each happen, and they are mutually exclusive. Explain why they can never be independent. (What does independence say P(A and B) must equal?)

> **Answer.** Independent means P(A and B) = P(A) × P(B). If both events are possible, that product is above 0. Mutually exclusive means P(A and B) = 0. Both cannot be true at once, so mutually exclusive events with positive probabilities are never independent.

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive. Write the check under it.

> **Answer** (one of many). Independent: sport 60, instrument 50, both 30 → sport only 30, instrument only 20, neither 20. Check: P(instrument | sport) = 30/60 = 50% = P(instrument) = 50/100. Any table with both = P(S) × P(I) × 100 works. Extension for a fast group: a second table where the events ARE mutually exclusive (sport 50, instrument 50, both 0, neither 0); the same check fails, 0/50 = 0% against P(instrument) = 50%.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 3 — checklist, roles, sketch

Can't happen together, so unrelated?

Poster checklist (this is the rubric)

- ☐ The joint probability P(Telepathy and Famous) from the table, and what it tells you
- ☐ The independence test, with its two numbers
- ☐ One verdict sentence that uses *mutually exclusive* AND *independent* correctly
- ☐ The proof from part (d), in two or three sentences
- ☐ Your 2x2 table (100 students) with the independence check written under it

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The probabilities, and the numbers in your 2x2 table.	The big table with the Telepathy row and the Famous column shaded, and your 2x2 table with its check underneath.	The verdict sentence and the proof.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: mutually exclusive | independent | given

Draft the poster sentences here.

Poster 4 — Double counted

DOK 3

Data

A survey of 2,500 adults asked about education level and where they get their news. One of these 2,500 adults is picked at random.

Group	Count
College graduates	693
Get news mainly from the internet	687
College graduates who get news from the internet	245
All adults surveyed	**2,500**

The claim you are judging

A blog wrote: "P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%. And college grads are no more likely than anyone else to get their news online: only 245/2500 = 9.8% of adults are grads who read news online."

Two mistakes are hiding in this claim. Find both.

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Draw a Venn diagram (or a two-way table) and fill in all four regions: both, grad only, internet only, neither. Check that they add to 2,500.

> **Answer.** Both 245 · grad only 693 − 245 = 448 · internet only 687 − 245 = 442 · neither 2,500 − 1,135 = 1,365. Sum 2,500.

b. Find the mistake in the 55.2%. Compute the correct P(grad or internet). Which people were counted twice?

> **Answer.** The 245 adults who are both were counted in 693 and again in 687. P(grad or internet) = (693 + 687 − 245)/2500 = 1135/2500 = 45.4%. (The blog's 1380/2500 = 55.2% is a union with double counting.)

c. The blog backs its second sentence with 9.8%. Is 9.8% a joint or a conditional probability? Which probability does "no more likely than anyone else" really depend on?

> **Answer.** 9.8% = 245/2500 = P(grad and internet), a joint probability out of all 2,500 adults. "No more likely than anyone else" compares grads with everyone, so it depends on the conditional P(internet | grad), set beside P(internet).

d. Test the second sentence with the right probability. Are the two events independent? Who is more likely to read news online?

> **Answer.** P(internet | grad) = 245/693 = 35.4% vs P(internet) = 687/2500 = 27.5%: not equal, not independent. Grads are MORE likely to read news online (35.4% vs 27.5% overall, and vs 442/1807 = 24.5% of non-grads). The blog's second sentence is false. (The product check agrees: P(grad) × P(internet) = 0.2772 × 0.2748 = 0.0762 vs P(both) = 245/2500 = 0.098.)

e. Build. Keep 693 college grads and 687 internet readers. How many people in "both" would independence predict? Compare that with 245. What does the gap tell you?

> **Answer.** both = 693 × 687 / 2500 = 190.44 people, about 190. The survey found 245, about 55 more than independence predicts, so the dependence is large, not rounding. (190.44 is not a whole number, so tables of these sizes can never be exactly independent.)

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 4 — checklist, roles, sketch

Double counted

Poster checklist (this is the rubric)

- ☐ Four regions filled and summing to 2,500 (both / grad only / internet only / neither)
- ☐ The people counted twice, named, and the corrected union as a fraction and a percent
- ☐ The blog's 9.8% named (joint or conditional?) and the probability it should have used
- ☐ The independence test with its two numbers, and a verdict naming who is more likely
- ☐ The "both" count independence would predict, compared with 245

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The four regions, the union, the test, the predicted "both" count. Show every subtraction.	The Venn diagram with the overlap shaded (or the 2x2 table), every region labeled with its count.	The double-count sentence, the sentence about the 9.8%, the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: union · given · independent

Draft the poster sentences here.

Poster 5 — Is the game fair?

DOK 3

Data

A carnival game costs $2 to play. You roll a fair six-sided die. Roll a 6: you win $10. Roll a 1: you win $3. Otherwise: you win nothing.

The claim you are judging

The sign on the booth: "A FAIR GAME! You win on 2 of every 6 rolls."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Let X = the player's net profit for one game (winnings minus the $2). List every value of X with its probability in a table. Check that the probabilities add to 1.

> **Answer.** X = +8 (roll 6) with 1/6 · +1 (roll 1) with 1/6 · −2 (roll 2, 3, 4, 5) with 4/6. Sum 6/6 = 1. Using $10, $3, $0 instead of net is the first trap.

b. Compute the mean μ_X and the standard deviation σ_X. Show the formula and the substitution; a calculator may do the arithmetic.

> **Answer.** $\mu_X = 8(1/6) + 1(1/6) + (-2)(4/6) = (8 + 1 - 8)/6 = 1/6 \approx \$0.17$. $\sigma_X^2 = (8 - 1/6)^2(1/6) + (1 - 1/6)^2(1/6) + (-2 - 1/6)^2(4/6) = 485/36 \approx 13.47$, so $\sigma_X \approx \$3.67$. Shortcut: $E(X^2) = (64 + 1 + 16)/6 = 13.5$, minus $\mu^2 = 1/36$.

c. Interpret μ_X and σ_X in context. Use "in the long run" and say "per game".

> **Answer.** In the long run, over many games, a player gains about 17 cents per game on average. A single game's net typically differs from that average by about $3.67.

d. Verdict. Is the game fair? Who does it favor? Why is "win on 2 of every 6 rolls" the wrong number for judging fairness?

> **Answer.** A fair game has $\mu_X = 0$, for both sides. Here $\mu_X = +\$0.17$, so the game favors the PLAYER: the booth loses about 17 cents per game in the long run. "2 of 6" is the chance of winning something on one roll (not a promise of two wins in six). It counts how OFTEN you win, not how MUCH: the $10 prize outweighs the four $2 losses.

e. Repair. Change ONE payout so that $\mu_X = 0$ exactly. Show the new table and the new mean.

> **Answer** (either). The 6 pays $9 (net +7): $\mu = (7 + 1 - 8)/6 = 0$. Or the 1 pays $2 (net 0): $\mu = (8 + 0 - 8)/6 = 0$. Raising the price to $13/6 \approx \$2.17$ also balances it (exactly only at $13/6$) but is not a payout change.

f. Simulate 20 games. On the TI-84: randInt(1,6,20) STO→ L1, then read the 20 rolls in STAT → EDIT. Use your table to turn each roll into its net, and find the average. Why does it differ from μ_X? What would 1,000 games do?

> **Answer.** Twenty-game averages vary a lot (their standard deviation is about $3.67/\sqrt{20} \approx \$0.82$), so most groups land somewhere between about −$1.50 and +$1.80. With 1,000 games the average settles near $0.17: the law of large numbers.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 5 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The distribution table of X = net profit, with probabilities that sum to 1
- ☐ μ_X and σ_X, each with its formula and substitution
- ☐ Both interpretations in context, with "in the long run" and "per game"
- ☐ A verdict naming who the game favors and why "2 of 6" is the wrong number
- ☐ The one-payout fix (new mean shown), and a dotplot of your 20 simulated games with their average marked

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The table, μ, σ, the fix. Show the formula before the numbers.	Runs the 20 games on the TI-84. The dotplot of the 20 nets, with their average and μ_X both marked.	The two interpretations and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____." **Also:** "In the long run, _____ per _____."

Your words: expected value | long run | net

Draft the poster sentences here.

Poster 6 (spare) — Rigged coin?

DOK 3

Data

You make the data: 20 flips of a fair coin on the TI-84, where 1 = heads. randInt(0,1,20) STO→ L1, then read the flips down L1 in STAT → EDIT. (randInt is under MATH → PROB.)

The claim you are judging

"I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

> **Answer.** One flip: a random digit 0 or 1, 1 = heads, each equally likely. One trial: 20 such digits, randInt(0,1,20). Record: the longest string of consecutive 1s in the trial.

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

> **Answer.** Typical values 3 to 5; a 6 or more appears in roughly 3 of 25 trials.

c. Estimate P(longest run is 6 or more) from your dotplot.

> **Answer.** (number of trials with a run of 6 or more)/25. Expect something near 0.12; anything from 0 to 0.24 is normal variation for 25 trials. If a group gets 0, pool the class's trials before the verdict.

d. Verdict. First decide: how rare would 6 in a row have to be before you believed the coin was rigged? Then compare your estimate with that bar. Is 6 in a row convincing evidence?

> **Answer.** No. A fair coin gives a run of 6 or more heads in about 12% of sets of 20 flips, roughly one set in eight. That is not rare, so it is not convincing evidence of rigging. It does not prove the coin is fair, either.

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

> **Answer.** A simulation gives a relative frequency, which varies from run to run; more trials make it settle closer to the exact value 0.1223 (law of large numbers). For reference: a run of 5 or more, 0.250; 7 or more, 0.058; 8 or more, 0.027.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 6 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The design in three sentences (flip, trial, what is recorded)
- ☐ 25 trials recorded in a table
- ☐ A dotplot of the 25 longest runs with a scale and a title
- ☐ The estimated probability as a fraction of 25 and a decimal
- ☐ The verdict, plus one sentence on what more trials would do

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
Runs the calculator, records the 25 longest runs, computes the fraction.	The dotplot, with a scale that covers every value you got and the "6 or more" region shaded.	The design sentences and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: simulation · trial · estimate

Draft the poster sentences here.

Exit ticket 1 — Tuesday (two per page, cut)

Exit ticket 1 · Tue 11/10 Name _____ Poster #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · union · independent · mutually exclusive · expected value.

- cut here -

Exit ticket 1 · Tue 11/10 Name _____ Poster #____

In your own words, no copying from the poster:
Our claim says _____.

_____

The number it really depends on is _____ = _____ (fraction, percent, or dollars).
So the claim is (supported / not supported / partly supported) because

_____

_____

_____

Use at least one of: given · union · independent · mutually exclusive · expected value.

Exit ticket 2 — Thursday (two per page, cut)

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. A classmate says: "P(Happy | Super Strength) and P(Super Strength | Happy) are equal, because both use the same 43 students." Is the classmate right? Show both as fractions and percents, and say why. (Super Strength total: 77. Happy total: 265.)

- cut here -

Exit ticket 2 · Thu 11/12 Name _____ Alone, on paper.

1. In the Super Status table, nobody who chose Telepathy wants to be Healthy (0 of 68). A classmate says: "So Healthy and Telepathy are independent." True or false? One sentence why, with a number. (P(Healthy) = 51/433.)

2. A classmate says: "P(Happy | Super Strength) and P(Super Strength | Happy) are equal, because both use the same 43 students." Is the classmate right? Show both as fractions and percents, and say why. (Super Strength total: 77. Happy total: 265.)

Run of show

Tue 11/10 · 09:55-11:00 · build (65 min)

Min	Students	Teacher
0-5	Do Now on whiteboards: "From the table, P(Rich | Fly)?" Boards up on the cue.	Scan. Ask one board: "Why 89 and not 433?" (22/89 = 24.7%.)
5-13	Three Reads on the projected table. Read 1: what is the story (no numbers). Read 2: list the quantities on your board. Read 3: what claims could someone make from this table?	The objectives, the EQ and the frame are already up on poster paper: point to them before Read 1. Collect three claims from boards; do not judge them yet.
13-17	Watch the example board.	Talk-track (3 min): the word that names the group, the denominator, verdict, repair, seven parts.
17-20	Groups form; cards and materials out; roles chosen and written on the card.	Hand each group its card. (Question cards go out Thursday.)
20-38	Solve parts (a)-(c) on scratch paper and whiteboards. No poster yet.	Circulate with the why questions. At minute 28, tier-1 hint card to any blank board; tier-2 only after that.
38-60	Build: title, the claim, the data, the probability, the visual. Draft (d) and (e). Poster 5: the calculator student runs the 20 simulated games now.	Check each group's verdict sentence before it goes on the board. Posters stay in the room.
60-65	Exit ticket 1, alone.	Collect. Sort by card for Thursday's defend round.

Thu 11/12 · 09:55-11:00 · observed: defend, repair, finish (65 min)

Min	Students	Teacher
0-6	Do Now on whiteboards: "True or false: P(Happy or Fly) = 265/433 + 89/433." Boards up.	Two why questions: "Who got counted twice, and how do you know?" "What would make the statement true?" (False: (265 + 89 − 48)/433 = 306/433 = 70.7%.)
6-8	Read today's job on the board.	Objectives and language objective re-read aloud; today's goal: defend, repair, finish.
8-26	Defend round: poster 1 with poster 2, poster 3 with poster 4; poster 5 presents to the teacher in round 1, then revises. 4 min: one group presents its verdict with the frame while the other asks from the question card. 2 min: the presenters revise their verdict sentence. Swap.	Time the rounds. Listen for the frame and the key words. Ask one open why question per group, then wait five seconds.
26-50	Repair and build parts of the card: (d) and (e). Finish the poster against the five-item checklist.	Circulate. Tier-2 hints only where a group is stuck on the repair. Check the checklist with each group.
50-56	Whole class at poster 3, on whiteboards: "Nobody who chose Telepathy wants to be Famous. Does knowing someone chose Telepathy change the chance they want to be Famous? Yes or no, plus one number." Boards up.	Wait five seconds, then boards up. The poster 3 group shows 0/68 against 15/433 = 3.5%. Then name the rule (the 10/29 promise): "Can't happen together rules the other one out, which changes its probability: dependence, not independence."
56-65	Exit ticket 2, alone, on paper.	Collect. These two items are the independent evidence for the day.

Run of show, Friday

Fri 11/13 · 08:55-09:50 · gallery walk (55 min)

Min	Students	Teacher
0-10	Final touches; posters up around the room.	Hand out walk sheets.
10-30	Walk: four rotations of 5 min, no docents; the poster must speak for itself (that is what the Writer's sentences are for). Start at the poster to your group's right. At each poster except your own, work out the one number on the walk sheet.	Signal rotations. Listen at the two persistent posters (2 and 3).
30-45	Debrief: hands for "which claim would have fooled you?"; two students read their sentence for posters 2 and 3.	Name the two rules: the given group is the denominator; mutually exclusive is not independent.
45-55	Hand in walk sheets; a fifth rotation for anyone who wants to revisit a poster.	Score with the grid (or after class); photos of the posters.

Observer card — Thursday 11/12

Lesson	Unit 2 claim posters, day 2 of 3: defend, repair, finish (DOK 3)
Plan	**Preparing (8 min).** Do Now on whiteboards, a packet problem: true or false, P(Happy or Fly) = 265/433 + 89/433. Boards up. Two open questions ("Who got counted twice, and how do you know?" "What would make the statement true?"). Objectives and the language objective re-read. **Interacting (42 min).** Defend round in partner groups: four minutes to present the verdict with the frame while the listening group asks from the question card (why that denominator, what would make the claim true, say it with "given"), two minutes to revise the verdict sentence, swap. Then the repair and build parts of each card: the repaired headline, the one-payout fair game, the designed 2x2 table, the count independence would predict. Finish the poster against the five must-haves. Tiered hint cards go only to a stuck group. **Extending (15 min).** Whole class at poster 3, on whiteboards: does knowing "Telepathy" change the chance of "Famous"? The promised example of events that are mutually exclusive but not independent, with the numbers. Exit ticket 2 alone on paper: a true-or-false on independence with a number, and a classmate's claim about two conditional probabilities to judge.
Sentence frame	"The claim says ___. The number it really depends on is ___ = ___. So the claim is (supported / partly supported / not supported) because ___."
Grouping	groups of three (one pair); partner groups for the defend round; alone for the exit ticket
Minutes	65

What the observer should count: open why questions asked; boards up at each cue; students who speak in the defend round; exit tickets collected.

Why questions (open; aim for more than half of all questions)

- Why is the denominator 89 and not 433?
- What would the claim need to say for your number to support it?
- Where did the 245 get counted twice?
- Why can't these two events be independent if they never happen together?
- What does +$0.17 promise a player, and what doesn't it promise?
- Which is the bigger group here, and why does that matter?
- If you only had the percents, could you still tell who is right?
- How would you convince someone who only looks at counts?
- What number in "both" would make these independent? Could that happen?
- Why does your simulation average differ from the mean?
- Can you say that again with the word "given"?
- Who does that sign fool, and who does it help?

Where each look-for lives on Thursday

Form item	Where
Objectives posted, content and language	The board page, up at 09:55; re-read at minute 6.
DOK level	DOK 3 on every card header and on the board: judge a claim, prove it, repair it.
Open questions, wait time	The why list; five seconds after each; boards up before anyone answers.
Language routine with frames	The verdict frame and the question cards in the defend round; word bank on the board.
Student-to-student discourse	Defend round, four minutes each way, every group speaks.
Whiteboards used	Do Now, repair scratch work, boards up at cues.
Visible independent work at the end	Exit ticket 2, alone, on paper, collected.
Scaffolds for multilingual learners	Word bank, sentence frames, tiered hint cards, the data table reprinted on every card, roles that give every student a job.
Evidence for feedback and adjustment	Exit tickets 1 and 2, the walk sheets and the scored posters, sorted by card.

Question cards (defend round) — one per group, cut

The listening group asks from its card. The presenting group answers with the frame. Nobody says "you're wrong"; you ask a question instead.

Question card

- Which word in the claim did you test?
- Show me the one number that decides it. Why that denominator?
- What would make the claim true?
- Say your verdict again with your key word.

Question card

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Question card

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Question card

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Question card

- Which word in the claim did you test?
- Show me the one number that decides it. Why that denominator?
- What would make the claim true?
- Say your verdict again with your key word.

Hint cards — tiered, hand out, never post

Tier 1 goes to a blank board at minute 28 on Tuesday. Tier 2 goes only to a group still stuck after tier 1, and only on the part they are stuck on. Frames come after thinking, not before.

Tier 1 (any card)

Find the word in the claim that names a group: *given, of, among, who, people*. Circle that group's TOTAL in the table. That total is your denominator. Now find the cell the claim is about and write the fraction.

Tier 2 · Card 1

Freeze Time row: 26 out of 101. Telepathy row: 19 out of 68. Which is bigger as a percent? Do all five rows the same way.

Tier 2 · Card 2

First sentence: you are inside the Famous COLUMN (15 people). Second sentence: inside the Fly ROW (89 people). Same cell, 5, on top of both.

Tier 2 · Card 3

Mutually exclusive asks: can both happen? (Look at the cell.) Independent asks: does knowing one CHANGE the chance of the other? Compare P(Famous) = 15/433 with P(Famous | Telepathy) = 0/68.

Tier 2 · Card 4

Draw two overlapping circles. Put 245 in the overlap FIRST. Grad-only = 693 − 245. Internet-only = 687 − 245. Neither = 2,500 minus everything inside the circles.

Tier 2 · Card 5

Net = winnings − 2. Rolls 2, 3, 4 and 5 all give the same net, so group them: probability 4/6. Mean = each net times its probability, added up.

Tier 2 · Card 6

MATH → PROB → randInt(0,1,20) STO→ L1 puts 20 flips in L1. Read down L1 in STAT → EDIT and count the longest string of 1s. Run it again for the next trial.

Rubric and scoring grid

Each card's five must-haves are the rubric. All five = E = 115. Three or four = P = 100. Fewer = I = 90. Any integer 90 to 115 is allowed per sheet (POSTER_WORK_TRACK_SPEC.md). Posters are 30% of Q2 Work for Period B; a missing poster is absent, never 0.

Card	M1	M2	M3	M4	M5	E / P / I	Score 90-115	Names
1 Who wants to be rich?								
2 The flipped headline								
3 Can't happen together								
4 Double counted								
5 Is the game fair?								
6 Rigged coin?								

Must-haves by card (M1 to M5)

Card	M1	M2	M3	M4	M5
1	Five Rich percents with fractions, row totals as denominators	Segmented bar chart: five 100% bars, four segments, key, percent axis	Verdict compares Freeze Time's percent with the other groups and the overall 23.6%	Repaired claim, true, with a percent	Association sentence comparing two bars
2	Both conditionals, denominators 15 and 89, percents	Two shaded tables showing the given groups	Verdict on each half; "most" vs 50%; the flip named	Repaired headline, two true sentences with numbers	Second flipped pair with both numbers and a why
3	Joint probability 0, "mutually exclusive"	Independence test with numbers (3.5% vs 0)	Verdict using both terms correctly	General argument (ME + possible means not independent)	A 2x2 table, independent but not mutually exclusive, with its check
4	Four regions summing to 2,500	Double count named; corrected union as fraction and percent	The 9.8% named as a joint; the conditional it should have used	Independence test, verdict, who is more likely	Predicted "both" count (about 190) against 245
5	Net distribution table, probabilities sum to 1	μ and σ with formula and substitution	Both interpretations, "long run", "per game"	Verdict: who it favors, why 2 of 6 is the wrong number	One-payout fix with $\mu = 0$, and the 20-game dotplot with its average beside μ
6	Design in three sentences	25 trials recorded	Dotplot with scale and title	Estimate as a fraction of 25 and a decimal	Verdict plus the more-trials sentence

Entering scores and grouping

Entering scores

File outside the repo (student names), then `node scripts/enter-poster.mjs <file>` to preview and `--apply` to write:
`unit=2 quarter=Q2 title=Unit 2 Poster (Claim, Proof, Repair) section=PeriodB`
`Real Name|115` or `Real Name|E` (E 115 / P 100 / I 90), one line per student.

Grouping grid (fill in before Tuesday)

Group	Card	Table Keeper	Visual	Writer
A				
B				
C				
D				
E (pair)	2		(Table Keeper)	

Answer key — Do Nows, exit tickets, the numbers, debrief notes

Do Nows (packet problems, whiteboards)

- **Tue:** P(Rich | Fly) = 22/89 = 24.7%. Denominator 89: inside the Fly row.
- **Thu:** False. P(Happy or Fly) = (265 + 89 − 48)/433 = 306/433 = 70.7%. The 48 students who are Happy AND Fly were counted twice in 354/433 = 81.8%.

Exit ticket 1 (Tue)

Full credit: the claim restated, the right number for that card (a conditional with the right denominator on cards 1-3, the corrected union and the conditional on card 4, the expected net in dollars on card 5, the estimated run probability on card 6), a verdict that matches the card's answer, one word from the bank used correctly. Sort by card; a group whose tickets disagree with each other gets the first visit Thursday.

Exit ticket 2 (Thu)

1. False. Healthy and Telepathy are mutually exclusive (0 of 68), not independent: P(Healthy) = 51/433 = 11.8% but P(Healthy | Telepathy) = 0/68 = 0%. Knowing Telepathy changes the chance.
2. No. P(Happy | Super Strength) = 43/77 = 55.8% is far bigger than P(Super Strength | Happy) = 43/265 = 16.2%. Same cell (43), different given group (77 vs 265).

The numbers, card by card

Card	Key values
1	P(Rich | power): Fly 22/89 = 24.7%, Freeze Time 26/101 = 25.7%, Invisibility 22/98 = 22.4%, Super Strength 13/77 = 16.9%, Telepathy 19/68 = 27.9%; P(Rich) = 102/433 = 23.6%. All five groups sit between 17% and 28%: Telepathy is highest, 2.2 points over Freeze Time, and no group stands out. Segmented bars (Famous / Happy / Healthy / Rich, %): Fly 5.6 / 53.9 / 15.7 / 24.7 · Freeze Time 3.0 / 62.4 / 8.9 / 25.7 · Invisibility 3.1 / 63.3 / 11.2 / 22.4 · Super Strength 5.2 / 55.8 / 22.1 / 16.9 · Telepathy 0.0 / 72.1 / 0.0 / 27.9.
2	P(Fly | Famous) = 5/15 = 33.3% (plurality, not majority); P(Famous | Fly) = 5/89 = 5.6%. Spare pairs: 43/77 = 55.8% vs 43/265 = 16.2%; 63/101 = 62.4% vs 63/265 = 23.8%; 19/68 = 27.9% vs 19/102 = 18.6%.
3	P(Telepathy and Famous) = 0; P(Famous) = 15/433 = 3.5% vs P(Famous | Telepathy) = 0; P(F) × P(T) = 0.0054 vs 0. Independent 2x2: 60 sport, 50 instrument, 30 both (check 30/60 = 50/100). Mutually exclusive 2x2: 50 / 50 / 0 both (check 0/50 ≠ 50/100).
4	Regions 245 / 448 / 442 / 1,365. Union 1135/2500 = 45.4% (not 55.2%). P(internet | grad) = 245/693 = 35.4% vs P(internet) = 27.5% vs non-grads 442/1807 = 24.5%. The blog's 9.8% = 245/2500 is the joint. Product 0.0762 vs joint 0.098. Independence would predict 190.44 in both.
5	X: +8 (1/6), +1 (1/6), −2 (4/6). μ = 1/6 ≈ \$0.17 for the player; σ^2 = 485/36 ≈ 13.47, σ ≈ \$3.67. Fixes: 6 pays \$9, or 1 pays \$2. SD of a 20-game average ≈ \$0.82.
6	P(run of 6+ heads in 20 fair flips) = 0.1223 (exact value, by computer); 5+: 0.250; 7+: 0.058; 8+: 0.027.

Debrief notes

- **The two persistent rules:** the word after "given" names the group whose total is the denominator (posters 1, 2, 4); mutually exclusive is the opposite of unrelated (poster 3, the 10/29 promise).
- **Card 1 nuance:** the newsletter's number is right and its reasoning is wrong. Accept "not supported" or "partly supported" when the sentence says the count is true and the percent does not stand out.
- **Card 5 nuance:** fair means $\mu_X = 0$, for both sides; the question is who an unfair game favors. The 20-game averages will scatter; put them on one class dotplot and the law of large numbers teaches itself.
- **Worksheet note (not for students):** the 2.7 follow-along's union examples print Super Status numbers (265 + 29 − 15; 115/433 + 29/433) that do not match the printed table (Super Strength is 77, Freeze Time 101, Happy and Super Strength 43). Every number in this kit uses the printed table; the worksheet blanks need a fix in a later pass.